

Work Smarter, Not Harder

An AI Training-Readiness Coach

Shiri, Lior, Elad

Team Members and Presenters



Push Today or Recover?

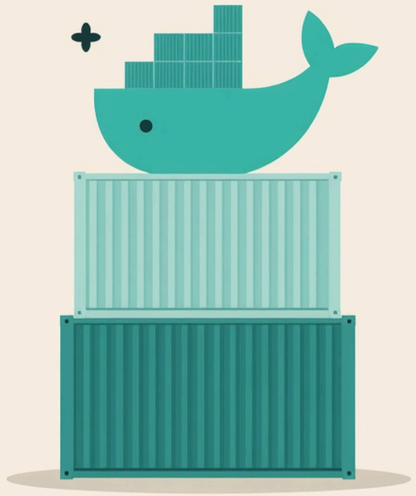
Athletes often struggle with **overtraining or undertraining**, leading to uncertain recovery. Our solution provides a quick 30-second morning check-in to signal **Ready, Moderate, or Rest** and offer personalized recommendations.



A Local AI Brain

Local Processing of Readiness Data

Our solution integrates a **local Random Forest classifier** and recommendation engine, ensuring no reliance on external APIs while providing accurate readiness assessments and personalized calorie targets.



Architecture Overview

Understanding our Docker-based container architecture

The system runs three Docker containers: web, AI, and MongoDB, with only the web container exposed for security. The AI runs behind a bounded process pool, and the model is baked into the image.



Online Forum Features

Our forum includes **seven core functionalities**: posts, comments, media sharing, likes/dislikes, direct messaging, notifications, and anti-abuse measures, all operating in real-time without refresh. An anonymity toggle enhances user privacy.



Security, Data, and Testing

Robust protocols for a secure application

Our application ensures **data integrity and security** through password hashing, two-step verification, and all five course test types — each one load-bearing and mutation-tested (not padding), including a real-Mongo integration suite and a cross-container E2E test that gate the build.

AI Job Queue and Scaling

Parallel predictions for efficient processing

2.86x

1.60x

Throughput increase from scaling

Process pool enhances processing efficiency significantly.

Scaling with replicas

Increased capacity improves handling of requests.

Commit to Live

It's live !!

app.worksmarternotharder.dev

The pipeline

push → test → build → deploy to Azure → HTTPS, with auto-rollback

Scan the QR to try it — and vote for us if you liked our idea